Faraday Rotation

$$\vec{E}(z,t) = E_{0x}\, e^{i(kz-\omega t)}\,\hat{x} + E_{0y}\, e^{i(kz-\omega t)}\,\hat{y}$$

$$\ddot{\vec{r}} = -\omega_0^2\,\vec{r} + \frac{q}{m}\,\dot{\vec{r}}\times\vec{B} + \frac{q}{m}\,\vec{E}$$

Electrons mex external static field $\vec{B}$ (this is not the field of the EM wave which we neglect in non-relativistic limit)

$$\begin{cases} \ddot{x} + \omega_0^2 x - \frac{qB}{m}\dot{y} = \frac{q}{m} E_{0x}\, e^{-i\omega t} \\ \ddot{y} + \omega_0^2 y + \frac{qB}{m}\dot{x} = \frac{q}{m} E_{0y}\, e^{-i\omega t} \end{cases}$$

$$\dot{\vec{r}}\times\vec{B} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ x & y & 0 \\ 0 & 0 & B \end{vmatrix} = \hat{x}\, yB - \hat{y}\, xB$$

$$x = x_0\, e^{-i\omega t}, \quad y = y_0\, e^{-i\omega t}$$

Assume, as usual, large wavelength $\Rightarrow$ can neglect z-dependence of $\vec{E}$.

$$(\omega_0^2-\omega^2)\, x_0 + i\omega\Omega_c\, y_0 = \frac{q E_{0x}}{m}$$

$$(\omega_0^2-\omega^2)\, y_0 - i\omega\Omega_c\, x_0 = \frac{q E_{0y}}{m}$$

$\Omega_c = \frac{qB}{m}$ cyclotron frequency

<u>Case of conductor</u>: $\omega_0 = 0$

$$\begin{pmatrix} -\omega^2 & i\omega\Omega_c \\ -i\omega\Omega_c & -\omega^2 \end{pmatrix}\begin{pmatrix} x_0 \\ y_0 \end{pmatrix} = \frac{q}{m}\begin{pmatrix} E_{0x} \\ E_{0y} \end{pmatrix}$$

$$\begin{pmatrix} x_0 \\ y_0 \end{pmatrix} = \begin{pmatrix} \frac{-1}{\omega^2-\Omega_c^2} & \frac{-i\Omega_c}{\omega(\omega^2-\Omega_c^2)} \\ \frac{i\Omega_c}{\omega(\omega^2-\Omega_c^2)} & \frac{-1}{\omega^2-\Omega_c^2} \end{pmatrix}\left(\frac{q}{m}\right)\begin{pmatrix} E_{0x} \\ E_{0y} \end{pmatrix}$$

For $\omega > \omega_p \gg \Omega_c$ (consider propagating wave)

$$\begin{pmatrix} x_0 \\ y_0 \end{pmatrix} = -\frac{q}{m\omega^2}\begin{pmatrix} 1 & -\frac{i\Omega_c}{\omega} \\ \frac{i\Omega_c}{\omega} & +1 \end{pmatrix}\begin{pmatrix} E_{0x} \\ E_{0y} \end{pmatrix}$$

$$\Rightarrow \quad x(t) = -\frac{q}{m\omega^2}\left(E_{0x}\, e^{-i\omega t} - \frac{i\Omega_c}{\omega} E_{0y}\, e^{-i\omega t}\right)$$

$$y(t) = -\frac{q}{m\omega^2}\left(E_{0y}\, e^{-i\omega t} + \frac{i\Omega_c}{\omega} E_{0x}\, e^{-i\omega t}\right)$$

$$\vec{j} = Nq\,\dot{\vec{r}}$$

$$\dot{x} = \frac{iq}{m\omega} E_{0x}\, e^{-i\omega t} + \frac{q\Omega_c}{m\omega^2} E_{0y}\, e^{-i\omega t} \qquad \frac{iq\Omega_c}{m\omega^3}(-i\omega)E_{0x}$$

$$\dot{y} = \frac{iq}{m\omega} E_{0y}\, e^{-i\omega t} - \frac{q\Omega_c}{m\omega^2} E_{0x}\, e^{-i\omega t}$$

$$j_x = \frac{iNq^2}{m\omega} E_{0x}\, e^{-i\omega t} + \frac{Nq^2\Omega_c}{m\omega^2} E_{0y}\, e^{-i\omega t}$$

$$j_y = -\frac{Nq^2\Omega_c}{m\omega^2} E_{0x}\, e^{-i\omega t} + \frac{iNq^2}{m\omega} E_{0y}\, e^{-i\omega t}$$

$$\vec{j} = \begin{pmatrix} \sigma_{xx} & \sigma_{xy} \\ \sigma_{yx} & \sigma_{yy} \end{pmatrix} \begin{pmatrix} E_{0x} \\ E_{0y} \end{pmatrix} e^{-i\omega t}$$

Conductivity tensor

$$\sigma_{ij} = \frac{iNq^2}{m\omega}\delta_{ij} + \frac{Nq^2}{m}\frac{\Omega_c}{\omega^2}\varepsilon_{ijz}$$

$$\Rightarrow \quad \varepsilon_{ij} = \varepsilon_0\,\delta_{ij} + \frac{i}{\omega}\sigma_{ij}$$

$$= \varepsilon_0\left(1 - \frac{\omega_p^2}{\omega^2}\right)\delta_{ij} + i\,\varepsilon_0\,\frac{\omega_p^2\,\Omega_c}{\omega^3}\,\varepsilon_{ijz}$$

$\Rightarrow$ EM waves in an anisotropic medium

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\vec{\nabla} \times \vec{H} = \frac{\partial \vec{D}}{\partial t} \quad \text{where } \vec{D} = \varepsilon \vec{E}, \ \varepsilon = \varepsilon_b + \frac{i\sigma}{\omega}$$

$$\vec{\nabla} \times (\vec{\nabla} \times \vec{E}) = -\nabla^2 \vec{E} + \underbrace{\vec{\nabla}(\vec{\nabla} \cdot \vec{E})}_{\emptyset} = -\frac{\partial}{\partial t}(\vec{\nabla} \times \vec{B})$$

$$\vec{\nabla} \times \vec{B} = \mu \vec{\nabla} \times \vec{H} = \mu \frac{\partial \vec{D}}{\partial t}$$

$$-\nabla^2 \vec{E} = -\mu \frac{\partial^2 \vec{D}}{\partial t^2} = -\mu \varepsilon \frac{\partial^2 \vec{E}}{\partial t^2} = (-\mu\varepsilon)(-\omega^2) \vec{E}(z) e^{-i\omega t}$$

For transverse wave $\vec{E}(z,t) = \vec{E}(z) e^{-i\omega t}$

$\hookrightarrow E_x(z)\hat{x} + E_y(z)\hat{y}$

$$\Rightarrow \frac{d^2}{dz^2}\begin{pmatrix} E_x \\ E_y \end{pmatrix} = -\frac{\omega^2}{c^2} \frac{\mu\varepsilon}{\mu_0 \varepsilon_0} \begin{pmatrix} E_x \\ E_y \end{pmatrix} = -\frac{\omega^2}{c^2}\left\{\left(1 - \frac{\omega_p^2}{\omega^2}\right)\delta_{ij} + \frac{i\omega_p^2 \Omega_c}{\omega^3} \varepsilon_{ijz}\right\}\begin{pmatrix} E_x \\ E_y \end{pmatrix}$$

$$\cancel{\mu\varepsilon = n^2 \mu_0 \varepsilon_0 = \frac{n^2}{c^2}}$$

Therefore for "vector of state" $\psi = \begin{pmatrix} E_x(z) \\ E_y(z) \end{pmatrix}$

$$\frac{d^2\psi}{dz^2} = \hat{M}\psi, \quad M = \begin{pmatrix} M_{xx} & M_{xy} \\ M_{yx} & M_{yy} \end{pmatrix}, \quad M_{yx} = -M_{xy}$$

Want to find eigenvectors $\hat{M}\psi = \lambda\psi$
(eigenstate of matrix M)

$$M_{xx} = -\frac{\omega^2}{c^2}\left(1 - \frac{\omega_p^2}{\omega^2}\right) = M_{yy}$$

$$M_{xy} = -\frac{\omega^2}{c^2} \cdot \frac{i\omega_p^2 \Omega_c}{\omega^3} = -M_{yx}$$

Therefore

$$\begin{cases} (M_{xx}-\lambda)\, E_x + M_{xy}\, E_y = 0 \\ M_{yx}\, E_x + (M_{yy}-\lambda) E_y = 0 \end{cases}$$

$$\det\begin{pmatrix} M_{xx}-\lambda & M_{xy} \\ M_{yx} & M_{yy}-\lambda \end{pmatrix} = 0 \Rightarrow \lambda^2 - \lambda\left(M_{xx}+M_{yy}\right) + \left(M_{xx}M_{yy} - M_{xy}\cdot M_{yx}\right) = 0$$

$$\lambda_\pm = \frac{M_{xx}+M_{yy}}{2} \pm \frac{1}{2}\sqrt{\left(M_{xx}+M_{yy}\right)^2 - 4\left(M_{xx}M_{yy} - M_{xy}M_{yx}\right)}$$

$$= \frac{M_{xx}+M_{yy}}{2} \pm \frac{1}{2}\sqrt{\left(M_{xx}-M_{yy}\right)^2 + 4M_{xy}M_{yx}}$$

$$(M_{xx}=M_{yy}) \quad = M_{xx} \pm \frac{1}{2}\cdot 2\sqrt{-M_{xy}^2} = M_{xx} \pm \sqrt{-M_{xy}^2} =$$

$$= -\frac{\omega^2}{c^2} \pm \sqrt{\left(\frac{\omega^2}{c^2}\right)^2 - (-i^2)\left(\frac{\omega_p^2\Omega_c}{\omega^3}\right)^2}$$

$$= -\frac{\omega^2}{c^2} \pm \frac{\omega^2}{c^2}\frac{\omega_p^2\Omega_c}{\omega^3}$$

$$= -\frac{\omega^2}{c^2} \pm \frac{\omega_p^2\Omega_c}{c^2\omega}$$

$$\frac{d^2\psi_+}{dz^2} = \lambda_+\psi_+$$

$$\psi_+(z) = \psi_+(0)\, e^{\pm i\sqrt{-\lambda_+}\, z}$$

(~~Same for λ_-~~)

$$-\lambda_\pm = \frac{\omega^2}{c^2}\left(1 \mp \frac{\omega_p^2\Omega_c}{\omega^3}\right)$$

$$\Rightarrow \sqrt{-\lambda_\pm} \cong \frac{\omega}{c}\left(1 \mp \frac{\omega_p^2\Omega_c}{2\omega^3}\right)$$

So that $\sqrt{-\lambda_\pm}$ is just wavevector $K_\pm$.

$$-\lambda_\pm = \frac{\omega^2}{c^2}\left(1 - \frac{\omega_p^2}{\omega^2}\right) \mp \frac{\omega^2}{c^2}\frac{\omega_p^2\Omega_c}{\omega^3}$$

$$K_\pm = \sqrt{-\lambda_\pm} = \frac{\omega}{c}\sqrt{n^2 \mp \frac{\omega_p^2\Omega_c}{\omega^3}} \approx \frac{\omega n}{c}\left(1 \mp \frac{\omega_p^2\Omega_c}{2\omega^3 n^2}\right)$$

$$n^2 = 1 - \frac{\omega_p^2}{\omega^2}$$

What are eigenvectors $\psi_\pm$ corresponding to $\lambda_\pm$?

$$M_{xx} - \lambda_\pm = \mp\sqrt{-M_{xy}^2}$$

$$\mp\sqrt{-M_{xy}^2}\, E_x^\pm + M_{xy}\, E_y^\pm = 0$$

$$\frac{E_x^\pm}{E_y^\pm} = \frac{\pm M_{xy}}{\sqrt{-M_{xy}^2}} = \pm(-i) = \mp i.$$

$$\Rightarrow \psi_+(z) = E_x^+(z)\begin{pmatrix}1\\ i\end{pmatrix}, \quad \psi_-(z) = E_x^-(z)\begin{pmatrix}1\\ -i\end{pmatrix}$$

$$\vec{E}^{(\pm)}(z,t) = \left(E_x(z)\,\hat{x} + \left(\pm i\, E_x(z)\right)\hat{y}\right)e^{-i\omega t}$$

$$= E_x(z)\left(\hat{x} \pm i\,\hat{y}\right)e^{-i\omega t}$$

$\Rightarrow$ the eigenvectors are EM waves of right/left circular polarization !!!

Kramers-Kronig properties of ε_{ij}

Observe $$\varepsilon_{ij}(\omega) = \varepsilon_0 \begin{pmatrix} 1 - \frac{\omega_p^2}{\omega^2} & i\,\frac{\omega_p^2 \Omega_c}{\omega^3} \\ -i\,\frac{\omega_p^2 \Omega_c}{\omega^3} & 1 - \frac{\omega_p^2}{\omega^2} \end{pmatrix}$$

$$\varepsilon_{ij}(\omega) = \varepsilon_0 n^2 \delta_{ij} + i\,\varepsilon_0 \frac{\omega_p^2 \Omega_c}{\omega^3} \varepsilon_{ijz}$$

From reality of $\varepsilon_{ij}(t-t')$ follows $\varepsilon_{ij}^*(\omega) = \varepsilon_{ij}(-\omega)$

<u>Onsager relation</u> : under time-reversal $t \to -t$
$\vec{B} \to -\vec{B}$ $(\vec{M} \to -\vec{M})$
$\vec{p} \to -\vec{p}$
$\vec{E} \to \vec{E}$

$$\varepsilon_{ij}(\omega)\Big|_B = \varepsilon_{ji}(-\omega)\Big|_{-B}$$

Proof: write $\varepsilon_{ij}(\omega) = a_{ij}(\omega) + i\, b_{ij}(\omega)$
$$= \frac{1}{2}\left(\varepsilon_{ij} + \varepsilon_{ji}^*\right) + i\,\frac{1}{2i}\left(\varepsilon_{ij} - \varepsilon_{ji}^*\right) = \mathrm{Re}[\varepsilon] + i\,\mathrm{Im}[\varepsilon].$$

where $a_{ij} = a_{ji}^*$ (hermitian) $\to$ (c.c. + transpose)
$b_{ij} = b_{ji}^*$ (antihermitian)

Absorbed power $\overline{P} = \frac{1}{2}\omega\, b_{ij}(\omega) E_i E_j^*$

does not change under this operation $\downarrow t \to -t$

$$\overline{P} = \frac{1}{2}(-\omega)\left(-b_{ij}(-\omega)\right) E_i E_j^*$$

$$\Rightarrow \quad b_{ij}(\omega)\Big|_B = -b_{ij}(-\omega)\Big|_{-B}$$

Kramers–Kronig

$$\varepsilon'(\Omega) - \varepsilon_0 = \frac{1}{\pi} P\int_{-\infty}^{\infty} d\omega \; \frac{\varepsilon''(\omega)}{\omega - \Omega}$$

$$a_{ij}(\Omega) - \varepsilon_0 \delta_{ij} = P\int \frac{d\omega}{\pi} \; \frac{b_{ij}(\omega)}{\omega - \Omega}$$

$$a_{ij}(-\Omega)\Big|_{-B} - \varepsilon_0 \delta_{ij} = P\int_{-\infty}^{\infty} \frac{d\omega}{\pi} \; \frac{b_{ij}(\omega)\big|_{-B}}{\omega + \Omega} = (\omega = -\omega') =$$

$$= P\int_{-\infty}^{\infty} \frac{d\omega'}{\pi} \; \frac{b_{ij}(-\omega')\big|_{-B}}{-\omega' + \Omega} =$$

$$= P\int \frac{d\omega}{\pi} \; \frac{b_{ij}(\omega)\big|_{B}}{\omega - \Omega} = a_{ij}(\Omega)\Big|_{B} - \varepsilon_0 \delta_{ij}$$

$\Rightarrow$ obtain that $a_{ij}(\omega)\big|_{B} = a_{ij}(-\omega)\big|_{-B}$

$$\varepsilon_{ij}(\omega)\big|_{B} = a_{ij}(\omega)\big|_{B} + i\, b_{ij}(\omega)\big|_{B} =$$

$$= a^*_{ji}(\omega)\big|_{B} + i\, b^*_{ji}(\omega)\big|_{B} \quad \{\text{hermicity}$$

$$= a^*_{ji}(-\omega)\big|_{-B} - i\, b^*_{ji}(-\omega)\big|_{-B} \quad \begin{cases}\text{oddness } \hat{b} \\ \text{evenness } \hat{a}\end{cases}$$

$$= \varepsilon^*_{ji}(-\omega)\big|_{-B} = \varepsilon_{ji}(\omega)\big|_{-B} \quad \{\text{reality}$$

Our Drude's form $\varepsilon_{ij}(\omega) = \varepsilon_0 n^2 \delta_{ij} + \varepsilon_0 i \dfrac{\omega_p^2 \Omega_c}{\omega^3} \varepsilon_{ijz}$

has this property as $\varepsilon_{jiz} = -\varepsilon_{ijz}$

and $\Omega_c\big|_{-B} = -\Omega_c\big|_{B}$.